

Assignment 2

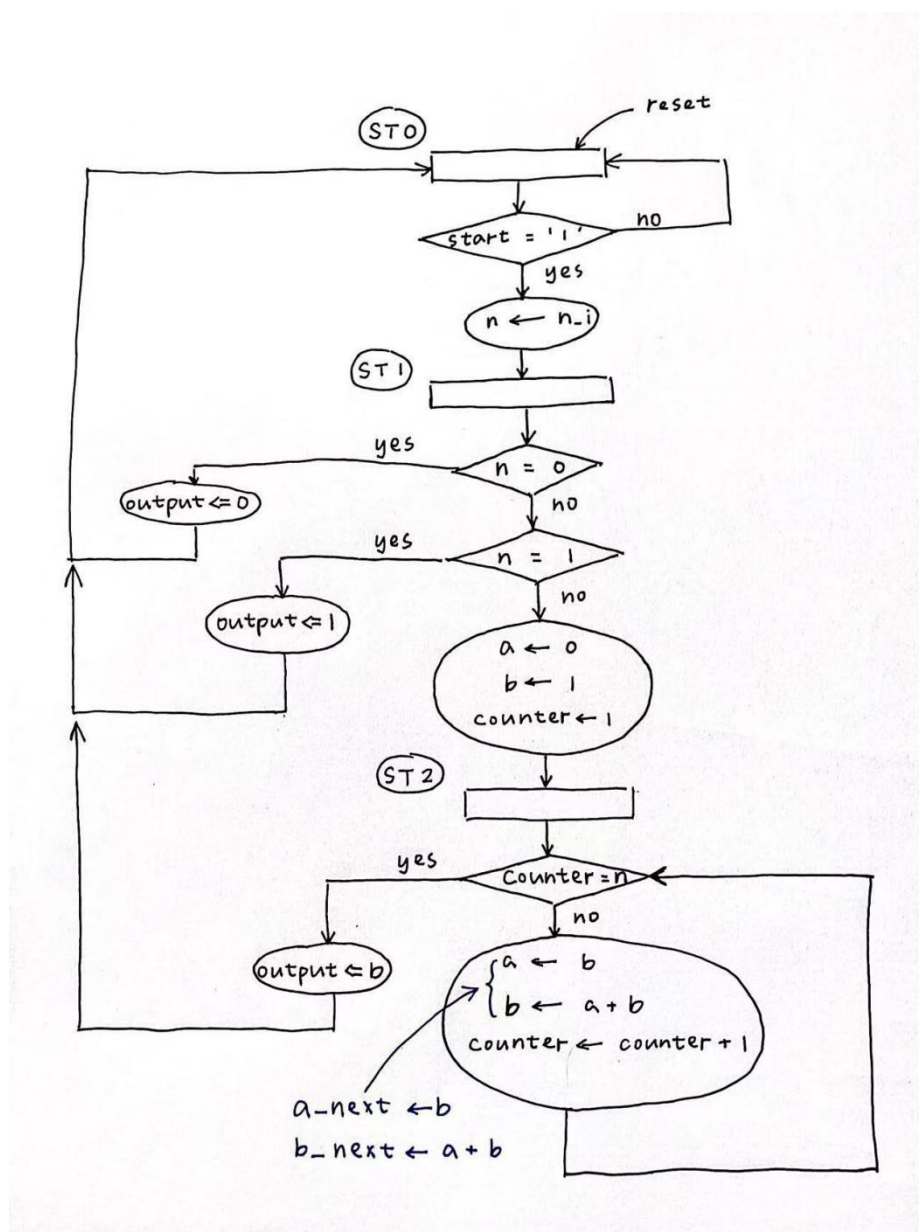
- 斐波那契数列

斐波那契数列定义如下：

$$fib(n) = \begin{cases} 0 & n = 0 \\ 1 & n = 1 \\ fib(n-1) + fib(n-2) & n > 1 \end{cases}$$

请用VHDL实现斐波那契数列计算。其中 n 为6位二进制输入，视为无符号整数。 $fib(63)=6557470319842$

算法状态机图



VHDL 代码

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity Fibonacci_sequence is
    Port ( n_i : in std_logic_vector(5 downto 0);
          clk : in STD_LOGIC;
          reset : in STD_LOGIC;
          start : in STD_LOGIC;
          output : out std_logic_vector(43 downto 0) );
end Fibonacci_sequence;

architecture FSM of Fibonacci_sequence is
    signal n, counter, counter_next : std_logic_vector(5 downto 0);
    signal a, b, a_next, b_next : std_logic_vector(43 downto 0);
    type S_Type is (ST0, ST1, ST2);
    signal state, next_state : S_Type;
begin
    clkproc: process(clk, reset) is
    begin
        if (reset= '1') then
            state <= ST0;
            a <= (others => '0');
            b <= x"000000000001";
            counter <= "000001"; -- counter 从 1 开始计数
        elsif (clk'event and clk= '1') then
            state <= next_state;
            a <= a_next;
            b <= b_next;
            counter <= counter_next;
        end if;
    end process clkproc;

    operproc: process (state, n_i, a, b, counter, start)is
    begin
        output <= (others => '0');
        a_next <= a;
        b_next <= b;
        counter_next <= counter;

        case state is
            when ST0 =>
```

```

        if (start = '1') then
            n <= n_i;
            next_state <= ST1;
        else
            next_state <= ST0;
        end if;
    when ST1 =>
        if (n = "000000") then
            output <= (others => '0');
            next_state <= ST0;
        elsif (n = "000001") then
            output <= x"000000000001";
            next_state <= ST0;
        else
            a_next <= (others => '0');
            b_next <= x"000000000001";
            counter_next <= "000001"; -- counter 从 1 开始计数
            next_state <= ST2;
        end if;
    when ST2 =>
        if (counter = n) then
            output <= b;
            next_state <= ST0;
        else
            a_next <= b;
            b_next <= a + b;
            counter_next <= counter + 1;
            next_state <= ST2;
        end if;
    end case;
end process operproc;
end FSM;

```

testbench 代码

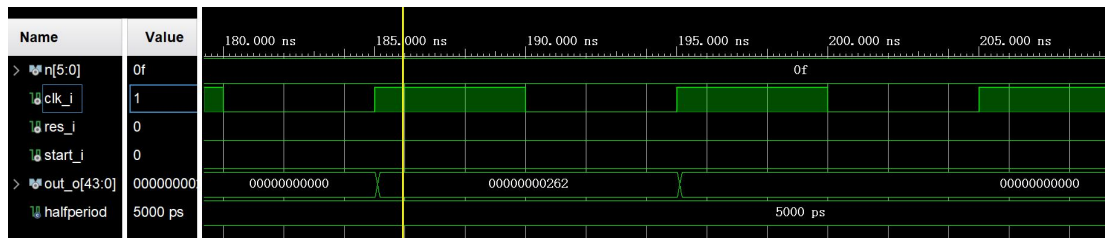
```

library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity Fibonacci_sequence_tb is
    -- Port ( );
end Fibonacci_sequence_tb;

architecture tb of Fibonacci_sequence_tb is

```

当输入的信号 n 为 0x0f（转换到十进制为 15）时，计算输出的值为 0x262（转换到十进制为 610）。

number	0	1	1	2	3	5	8	13	21	34	55	89	144	233	377	610
n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

在上面的几次测试中，输出的值都符合斐波拉契数列的数字规律，说明电路功能正常。